
Problem 1. Let G and J be groups and let $\varphi : G \rightarrow H$ be a group epimorphism (surjective homomorphism)

- (a) Prove that if J is abelian, then any subgroup of G that contains $\ker(\varphi)$ is a normal subgroup of G .

Proof. Note that the first isomorphism theorem says that

$$G/\ker(\varphi) \cong \varphi(G)$$

Also, $\varphi(G) = J$ since φ is surjective. So, $G/\ker(\varphi) \cong J$ and hence, $G/\ker(\varphi)$ is abelian. Since J is abelian, every subgroup of $G/\ker(\varphi)$ is normal. Hence, by the 4th iso theorem every subgroup of G containing $\ker(\varphi)$ is also normal. ■

- (b) Suppose $K \trianglelefteq J$. Prove that there exists $H \trianglelefteq G$ such that $\ker(\varphi) \subseteq H$ and G/H is isomorphic to J/K .

Proof. Suppose $K \trianglelefteq J$. We will prove that our candidate for H will be $\varphi^{-1}(K) \leq G$. Indeed, we can see that $\varphi^{-1}(K)$ is a subgroup of G from homework 2. Now, our goal is to show that $\varphi^{-1} \trianglelefteq G$. It suffices to show that for all $g \in G$,

$$g\varphi^{-1}(K)g^{-1} \subseteq \varphi^{-1}(K).$$

To this end, let $g \in G$ and let $x \in g\varphi^{-1}(K)g^{-1}$. Then we have

$$x = gzg^{-1} \text{ for some } z \in \varphi^{-1}(K).$$

Since $z \in \varphi^{-1}(K)$, we have $\varphi(z) \in K$. Then

$$\begin{aligned} \varphi(x) &= \varphi(gzg^{-1}) = \varphi(g)\varphi(z)\varphi(g^{-1}) \\ &= \varphi(g)\varphi(z)(\varphi(g))^{-1} \\ &= j'\varphi(z)(j')^{-1}. \end{aligned}$$

Note that because $K \trianglelefteq J$, it follows that for all $j \in J$,

$$jKj^{-1} \subseteq K.$$

In particular, for $j = j'$ and since $\varphi(z) \in K$, we have that $j'\varphi(z)(j')^{-1} \in K$. This tells us that $\varphi(x) \in K$ and so $x \in \varphi^{-1}(K)$. Thus, we see that, indeed, $\varphi^{-1}(K) \trianglelefteq G$. From part (a), we can see that $\ker(\varphi) \subseteq \varphi^{-1}(K)$. Hence, we can see that our candidate must be $H = \varphi^{-1}(K)$. Now, our goal is to show that $G/H \cong J/K$. We will do this through the First Isomorphism Theorem. Define the mapping

$$\psi : G \rightarrow J/K \text{ by } g \mapsto \varphi(g)K.$$

Indeed, we can see that

$$\begin{aligned} \ker(\psi) &= \{g \in G : \psi(g) = 1_H K\} \\ &= \{g \in G : \varphi(g)K = 1_H K\} \\ &= \{g \in G : \varphi(g) \in K\} \\ &= \varphi^{-1}(K). \end{aligned}$$

Hence, we have $\ker(\psi) = \varphi^{-1}(K) = H$. Because φ is a surjective map, it also follows from our definition of ψ that ψ must also be surjective. Indeed, for any $h \in H$, there exists an $g \in G$ such that $\varphi(g) = h$. Hence, we have

$$hK = \varphi(g)K = \psi(g)$$

and so $\psi(G) = J/K$. Using the First Isomorphism Theorem, we can see that

$$G/\ker(\psi) \cong \psi(G) \iff G/H \cong J/K$$

where $H = \varphi^{-1}(K)$. ■

Problem 2. Suppose that $\pi : \mathbb{Z}_{30} \rightarrow G$ is an onto homomorphism where G is a group of order 5. Determine, with proof, the kernel of φ . ■

Proof. Observe, by the First Isomorphism Theorem, (because π is an onto homomorphism) that

$$\mathbb{Z}_{30}/\ker(\pi) \cong \pi(\mathbb{Z}_{30}) \iff \mathbb{Z}_{30}/\ker(\pi) \cong G.$$

This tells us that, since G and $\mathbb{Z}_{30}$ are finite groups, it follows that

$$|\mathbb{Z}_{30}/\ker(\pi)| = |G| \iff \frac{|\mathbb{Z}_{30}|}{|\ker(\pi)|} = |G|.$$

Since $|G| = 5$ and $|\mathbb{Z}_{30}| = 30$, we have from the above that $|\ker(\varphi)| = 6$. This tells us that there are six elements of $\mathbb{Z}_{30}$ that gets mapped to the identity 1_G . Since the divisors of 30 gives us unique cyclic subgroups of $\mathbb{Z}_{30}$, all we need to do is find the subgroup of the same order as $\ker(\varphi)$. Indeed, this subgroup of order 6 must be the cyclic subgroup generated by $\langle \bar{5} \rangle$. That is, $\ker(\pi) = \langle \bar{5} \rangle = \{\bar{0}, \bar{5}, \bar{10}, \bar{15}, \bar{20}, \bar{25}\}$. ■

Problem 3. Suppose that $|G| = p^2q$ where p and q are primes with $p > q$. Show that if G has a subgroup H of order p^2 , then H is the unique subgroup of order p^2 in G . Deduce that G/H is cyclic. ■

Proof. ■

Problem 4. Suppose that H and K are subgroups of finite index in a possibly infinite group G with $|G : H| = m$ and $|G : K| = n$. Prove that $\text{lcm}(m, n) \leq |G : H \cap K| \leq mn$. Deduce that $|G : H \cap K| = mn$ when m and n are relatively prime. ■

Proof. ■

Problem 5. Prove Theorem 16 and corollary 17 from section 3.3 of the textbook. ■

Theorem (The First Isomorphism Theorem). If $\varphi : G \rightarrow H$ is a homomorphism of groups, then $\ker(\varphi) \trianglelefteq G$ and $G/\ker(\varphi) \cong \varphi(G)$.

Proof. Define $\mu : G/\ker(\varphi) \rightarrow \varphi(G)$ by $\mu(gK) = \varphi(g)$ where $K = \ker(\varphi)$. To see that K is normal, let $g \in G$ and let $y \in gKg^{-1}$. Then $y = gkg^{-1}$ for some $k \in \ker(\varphi)$. Then $\varphi(k) = 1_H$ and so

$$\begin{aligned} \varphi(y) &= \varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g^{-1}) \\ &= \varphi(g)\varphi(k)(\varphi(g))^{-1} \\ &= \varphi(g)1_H(\varphi(g))^{-1} \\ &= \varphi(g)(\varphi(g))^{-1} \\ &= 1_H. \end{aligned}$$

Hence, it follows that $y \in \ker(\varphi)$ and so, $K = \ker(\varphi)$ must be a normal subgroup. In what follows,

we will show that μ is isomorphism. First, we want to make sure that μ is well-defined. To this end, suppose $g_1K = g_2K$ where $g_1, g_2 \in G$. This is true if and only if $g_1^{-1}g_2 \in \ker(\varphi)$. This implies that $\varphi(g_1^{-1})\varphi(g_2) = 1_G$ which is true if and only if

$$(\varphi(g_1))^{-1}\varphi(g_2) = 1_G \iff \varphi(g_2) = \varphi(g_1) \iff \mu(g_2K) = \mu(g_1K).$$

This tells us that μ is well-defined and also that μ is injective. Next, we will show that μ is a homomorphism. Indeed, for all $g_1K, g_2K \in G/K$, we have that (since K is a normal subgroup)

$$\begin{aligned} \mu(g_1K g_2K) &= \mu(g_1g_2K) \\ &= \varphi(g_1g_2) \\ &= (\varphi(g_1)\varphi(g_2)) \\ &= \varphi(g_1)\varphi(g_2) \\ &= \mu(g_1K)\mu(g_2K). \end{aligned}$$

Lastly, we show that μ is surjective. It is clear from the definition that this is the case. Indeed, for any $y \in \varphi(G)$, it follows that $\varphi(g) = y$ for some $g \in G$ and so we have

$$\mu(gK) = \varphi(g) = y.$$

Hence, we see from the above argument that μ must be an isomorphism and so we conclude that

$$G/\ker(\varphi) \cong \varphi(G).$$

■

Corollary (17). Let $\varphi : G \rightarrow H$ be a homomorphism of groups. Then

- (1) φ is injective if and only if $\ker(\varphi) = 1$.
- (2) $|G : \ker(\varphi)| = |\varphi(G)|$.

Proof. First, we show (1). Since $\varphi : G \rightarrow H$ is a homomorphism, it follows from the First Isomorphism Theorem that

$$G/\ker(\varphi) \cong \varphi(G).$$

If $\ker(\varphi) = \{1_G\}$, then

$$G/\ker(\varphi) = G/\{1_G\} \cong G.$$

This holds if and only if

$$G \cong \varphi(G).$$

which is true if and only if φ is injective. Next, we show (2). Indeed, we can see from the First Isomorphism Theorem that (since φ is a homomorphism)

$$\begin{aligned} G/\ker(\varphi) \cong \varphi(G) &\iff |G/\ker(\varphi)| = |\varphi(G)| \\ &\iff |G : \ker(\varphi)| = |\varphi(G)|. \end{aligned}$$

■

Problem 6. Let G be a group with a normal subgroup N . Prove that there is a bijection from the subgroups of G containing N onto the subgroup of G/N . ■

Proof. ■